How the System Works

Physics, mathematics, system design and experiment results

GNSS Based Orbit Determination and Trajectory Prediction · Test satellite: ESA Sentinel-3A

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Index

How the system works: physics, mathematics, design and results

1. The problem in one page
2. The satellite and its orbit
3. Physics of the orbit: the force model
4. Physics of the measurements: GPS code and carrier phase
5. Cycle slips and data editing
6. Mathematics of the orbit determination
7. Mathematics of the prediction
8. System design
9. Experiments
10. Interpreting the results
11. Limitations and next steps
12. Summary

How the system works: physics, mathematics, design and results

This document explains the physics, the mathematics, the software design and the experimental results of the GNSS-based orbit determination and trajectory prediction tool. It is self-contained: each section starts from first principles and ends with the exact way the program implements it, with the setting names and the numbers it uses.

The test satellite throughout is **ESA Sentinel-3A**. The experiments cover **61 daily cut-offs from 3 September to 2 November 2024**, each at 05:59:42 UTC. Every forecast is scored against ESA's precise orbit product (**POEORB**), which is accurate to a few centimetres and is used for scoring only, never as an input.

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1. The problem in one page

A satellite in low Earth orbit (LEO) carries a GPS receiver. Every few seconds it records the signals of the GPS satellites in view. From those recordings (the **GNSSRD** files, in RINEX format) the program must:

1. **Determine the orbit (OD):** find the satellite's position and velocity at a chosen moment, the **cut-off**, using only data recorded up to that moment.
2. **Predict the trajectory:** integrate the equations of motion forward from the cut-off for up to 7 days (or longer) and write the result as a standard CCSDS OEM file.

Two rules govern every run:

- **Cut-off safety.** Nothing published or measured after the cut-off may be used: not GPS products, not Earth-orientation values, not solar or geomagnetic indices. A forecast that peeks at the future would look good in tests and fail in operations.
- **Scoring is separate.** POEORB is read only after the run, to measure the error.

The answer quality depends on two very different things:

Phase	What limits accuracy	Typical size
Orbit determination (up to the cut-off)	measurement modelling: clocks, antennas, ambiguities, cycle slips	1–2 m
Prediction (after the cut-off)	force modelling, above all **atmospheric drag** and future space weather	40 m after 1 day, 0.4–10 km after 3–7 days

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2. The satellite and its orbit

Property	Sentinel-3A value	Consequence
Altitude	about 814 km	thin but not negligible atmosphere
Inclination	98.65°, sun-synchronous	passes over the poles; constant local solar time
Orbital speed	$v = \sqrt{\mu/a} \approx 7.44$ km/s	1 ms of timing error = 7.4 m along-track
Period	$T = 2\pi\sqrt{a^3/\mu} \approx 101$ min	about 14.3 orbits per day
Mass	1130 kg	—
Drag / radiation area	8 m² (profile values)	area-to-mass ratio 0.0071 m²/kg
GPS antenna (`ANTENNA: DELTA`)	(2.881, −0.190, −0.794) m in the body frame	must be referred to the centre of mass

Property	Sentinel-3A value	Consequence
Centre of mass	(1.489, 0.217, 0.009) m	antenna lever arm about 1.66 m

Here $\mu = 3.986\,004\,418 \times 10^{14}$ m³/s² is Earth's gravitational parameter and $a \approx 6\,378 + 814$ km is the orbit radius.

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3. Physics of the orbit: the force model

3.1 Equation of motion

In an inertial frame (GCRF) the satellite's acceleration is the sum of every force per unit mass:

$$ \ddot{\mathbf r} = -\frac{\mu}{r^3}\mathbf r

- \mathbf a_{\text{geopotential}}
- \mathbf a_{\text{Sun}} + \mathbf a_{\text{Moon}}
- \mathbf a_{\text{solid tides}} + \mathbf a_{\text{ocean tides}}
- \mathbf a_{\text{drag}}
- \mathbf a_{\text{SRP}} + \mathbf a_{\text{albedo/IR}}
- \mathbf a_{\text{relativity}} $$

Each term is computed by an Orekit force model. Their sizes at Sentinel-3A's altitude explain where the errors come from:

Force	Orekit class	Approximate size (m/s²)	Modelled to
Central gravity μ/r^2	`HolmesFeatherstoneAttractionModel`	7.7	exact
Earth oblateness (J_2)	same, EGM96 70×70	1×10^{-2}	≪ 1 %
Higher harmonics	same	10^{-4} – 10^{-6}	truncated at degree 70
Moon	`ThirdBodyAttraction` + DE440	1×10^{-6}	exact
Sun	same	6×10^{-7}	exact
Solid-Earth tides	`SolidTides` (IERS 2010)	$\sim 10^{-7}$	good
Ocean tides	`OceanTides` (FES2004, degree 4)	$\sim 10^{-8}$	8 main constituents

Force	Orekit class	Approximate size (m/s²)	Modelled to
Atmospheric drag	`DragForce` + `NRLMSISE00`	4×10^{-9} – 2×10^{-8}	**density error 20–100 %**
Solar radiation pressure	`SolarRadiationPressure`	4×10^{-8}	cannonball, $C_r = 1.3$
Earth albedo + infrared	`KnockeRediffusedForceModel`	$\sim10^{-9}$	good
Relativity	Schwarzschild + `LenseThirringRelativity`	$\sim10^{-8}$	exact

Drag is small compared with gravity but it is the term we know worst. Section 3.4 shows that it alone explains almost all of the forecast error.

3.2 Gravity

Earth's gravity potential is expanded in spherical harmonics:

$$U(r,\phi,\lambda) = \frac{\mu}{r}\left[1 + \sum_{n=2}^{N}\left(\frac{R_E}{r}\right)^n \sum_{m=0}^{n} \bar{P}_{nm}(\sin\phi)\left(\bar{C}_{nm}\cos m\lambda + \bar{S}_{nm}\sin m\lambda\right)\right]$$

with ϕ, λ the latitude and longitude in the Earth-fixed frame (ITRF), $\bar{P}_{nm}$ the normalised Legendre functions, and $\bar{C}_{nm}, \bar{S}_{nm}$ the EGM96 coefficients (`forces.gravity.degree` = `order` = 70). The acceleration is ∇U, rotated from ITRF into GCRF with the IERS 2010 conventions and the Earth-orientation parameters (EOP). Solid and ocean tides add small time-varying corrections $\Delta\bar{C}_{nm}(t), \Delta\bar{S}_{nm}(t)$.

3.3 Third bodies

The Sun and Moon pull on both the satellite and Earth. The orbit only feels the difference:

$$\mathbf{a}_b = \mu_b\left(\frac{\mathbf{r}_b - \mathbf{r}}{|\mathbf{r}_b - \mathbf{r}|^3} - \frac{\mathbf{r}_b}{|\mathbf{r}_b|^3}\right)$$

Their positions $\mathbf{r}_b$ come from the JPL **DE440** ephemeris, which the program requires.

3.4 Atmospheric drag: the dominant forecast error

The atmosphere moves with Earth, so the satellite feels a relative velocity $\mathbf{v}_r = \mathbf{v} - \boldsymbol{\omega}_E \times \mathbf{r}$, and

$$\mathbf{a}_{\text{drag}} = -\tfrac{1}{2}\rho(\mathbf{r},t)\frac{C_D A}{m}|\mathbf{v}_r|\,\mathbf{v}_r\,.$$

With $C_D = 2.2$, $A = 8$ m² and $m = 1130$ kg, $\tfrac{1}{2}C_D(A/m)v^2 \approx 4.3\times10^5$ m²/s²·(kg/m³)⁻¹. The density at 814 km, near the 2024 solar maximum, is of order 10^{-14} to 5×10^{-14} kg/m³, so drag is between about 4×10^{-9} and 2×10^{-8} m/s².

The density model. ρ comes from NRLMSISE-00, an empirical model driven by three space- weather inputs:

- $F_{10.7}$, the daily solar radio flux (a proxy for the extreme-ultraviolet heating of the upper atmosphere);
- $\bar{F}_{10.7}$, its 81-day mean;
- A_p, the geomagnetic index (heating by the solar wind; it jumps during storms).

After the cut-off nobody knows these values, so the program forecasts them (`ForecastSpaceWeatherProvider`):

- $F_{10.7}$ and $\bar{F}_{10.7}$ are held at their cut-off values (persistence);
- A_p is set to a scenario value, `weather.forecastAp` = 5 (quiet geomagnetic conditions).

This is the central physical limitation: a storm after the cut-off can raise the density several-fold, and no method that respects the cut-off can see it coming.

Why a small drag error becomes a large along-track error. Consider a constant error δf in the along-track acceleration. Gauss's equations give the change of the semi-major axis a and of the mean motion $n = \sqrt{\mu/a^3}$:

$$\dot{\delta a} = \frac{2}{n}\delta f, \qquad \delta n = -\frac{3}{2}\frac{n}{a}\delta a \;\Rightarrow\; \delta s(t) = a\int_0^t \delta n\, dt' = -\frac{3}{2}\delta f\, t^2 .$$

The along-track error grows with the **square of time**. The along-track component is 99.5 % of the total error after one day in the campaign (median of $|e_{\text{along}}|/|e_{3D}|$ over 59 scored days), so we can invert this formula on the measured medians to get the *equivalent* acceleration error:

Horizon	Before: δf (m/s²)	Final: δf (m/s²)
+1 day	8.5×10^{-9}	3.8×10^{-9}
+3 days	1.0×10^{-8}	4.4×10^{-9}
+7 days	1.5×10^{-8}	1.9×10^{-8}

Three conclusions follow:

1. Up to 3 days the error really is quadratic (nearly constant δf): it is a drag-level acceleration error.
2. Estimating C_D (step 1, section 9) more than halves δf, because C_D absorbs the density model's bias over the fitting arc.
3. By 7 days δf grows again: the space weather drifts away from the persisted values, and a C_D fitted to half a day of atmosphere no longer holds.

3.5 Radiation pressure

Sunlight carries momentum. With the solar pressure $P_\odot = 4.56 \times 10^{-6}$ N/m² at 1 AU:

$$\mathbf{a}_{\mathrm{SRP}} = -\nu P_\odot C_R \frac{A}{m} \left(\frac{1\,\mathrm{AU}}{|\mathbf{r}_\odot - \mathbf{r}|}\right)^2 \hat{\mathbf{s}},$$

where $\hat{\mathbf{s}}$ points to the Sun and $\nu \in [0, 1]$ is the fraction of the Sun visible (0 in Earth's shadow, between 0 and 1 in the penumbra). Earth's reflected sunlight (albedo) and its own infrared emission add a smaller push, modelled with Knocke's rediffused-radiation model.

3.6 Relativity

General relativity adds a correction to the central acceleration of order $(v/c)^2 \cdot \mu/r^2$, a few times $6 \times 10^{-10} \times 7.7$ m/s², that is about 10^{-8} m/s² (Schwarzschild term), plus the much smaller frame-dragging (Lense–Thirring) term. Both are switched on (`forces.relativity`).

3.7 Attitude

The orientation of the satellite matters twice: it places the GPS antenna (1.66 m from the centre of mass) and it orients the panels for drag and radiation. Sentinel-3A flies **yaw-steered nadir pointing**: the body points at the ground below and yaws continuously so the ground speed lies in the body's x–z plane. The program uses Orekit's `YawCompensation(NadirPointing)` wrapped in `CpodBodyFrame`, a fixed 180° turn about z that maps Orekit's convention to the mission's. Against the measured attitude quaternions (PROQUA files) the mean difference is **0.005°** (the previous approximation was 2.5° off, which moved the antenna by about 7 cm).

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4. Physics of the measurements: GPS code and carrier phase

4.1 What the receiver measures

Each GPS satellite transmits on two frequencies: $f_1 = 1575.42$ MHz (L1, wavelength $\lambda_1 = 19.0$ cm) and $f_2 = 1227.60$ MHz (L2, 24.4 cm). For each satellite and each epoch the receiver records two kinds of measurement on each frequency.

Code (pseudorange) P_i. The time the signal took to arrive, from the code modulated on it, times c. It is unambiguous but noisy (decimetres to metres).

Carrier phase L_i. The receiver counts the cycles of the carrier wave itself. It is about 100 times more precise (millimetres), but it starts with an unknown whole number of cycles N_i, the **ambiguity**, which stays constant only while the receiver keeps lock.

The two observation equations, in metres, for a GPS satellite s and the receiver r are:

$$\begin{aligned} P_i &= \rho + c(\delta t_r - \delta t^s) + I_i + \delta_{\mathrm{rel}} + \delta_{\mathrm{Shapiro}} + \varepsilon_{P_i} \\ L_i &= \rho + c(\delta t_r - \delta t^s) - I_i + \delta_{\mathrm{rel}} + \delta_{\mathrm{Shapiro}} + \lambda_i N_i + \lambda_i \omega + \varepsilon_{L_i} \end{aligned}$$

Symbol	Meaning
ρ	geometric distance between the antennas, including light time
δt_r, δt^s	receiver clock and GPS-satellite clock offsets
$I_i = 40.3\,\mathrm{TEC}/f_i^2$	ionospheric delay (code is delayed, phase advanced)
$\delta_{\mathrm{rel}} = -2\,\mathbf{r}^s \cdot \mathbf{v}^s / c$	the GPS clock's relativistic periodic term
$\delta_{\mathrm{Shapiro}}$	delay of light in Earth's gravity field
$\lambda_i N_i$	the phase ambiguity
$\lambda_i \omega$	phase wind-up
ε	noise and multipath

There is no troposphere term: Sentinel-3A is above it.

4.2 Removing the ionosphere

The ionospheric delay depends on $1/f^2$, so a combination of the two frequencies cancels it exactly to first order. This is the **ionosphere-free (IF)** combination:

$$P_{IF} = \frac{f_1^2 P_1 - f_2^2 P_2}{f_1^2 - f_2^2} = 2.546\,P_1 - 1.546\,P_2\,, \qquad L_{IF} = 2.546\,L_1 - 1.546\,L_2\,.$$

The price is noise: $\sqrt{2.546^2 + 1.546^2} \approx 3$ times the single-frequency noise. The program uses P_{IF} and L_{IF} in metres (Orekit's `OneWayGNSSRange` and `OneWayGNSSPhase`, the latter with a nominal wavelength of 1 m so the ambiguity is in metres). The IF ambiguity $B_{IF} = 2.546\,\lambda_1 N_1 - 1.546\,\lambda_2 N_2$ is no longer an integer number of cycles, so it is estimated as a real number (a **float ambiguity**).

4.3 Light time, Sagnac, Shapiro

The signal leaves the GPS satellite at $t - \tau$ and arrives at t. The distance is found by solving, in the inertial frame,

$$c\,\tau = |\mathbf{r}_r(t) - \mathbf{r}^s(t - \tau)|,$$

by iteration (three iterations reach 1 µm). Doing it in the inertial frame automatically includes the **Sagnac effect**, the rotation of the Earth-fixed frame during the 70–90 ms of flight (up to about 30 m). The **Shapiro delay** is

$$\delta_{\mathrm{Shapiro}} = \frac{2\mu}{c^2} \ln \frac{r^s + r_r + \rho}{r^s + r_r - \rho} \quad \text{(a few cm).}$$

4.4 Antennas

The measurements are made at the antennas' **phase centres**, not at the centres of mass.

- **On Sentinel-3A** the antenna is 1.66 m from the centre of mass. The offset vector (DELTA minus centre of mass) is rotated by the attitude of section 3.7 and added to the position. Without the centre-of-mass correction the orbit is 1.9 m off along-track.
- **On each GPS satellite** the transmitting antenna sits up to about 2.5 m from the centre of mass, and its phase centre varies with the direction of transmission. The program reads the IGS values (`igs20.atx`, ANTEX format), combines the L1 and L2 values into IF values with the same weights as section 4.2, and rotates them with each GPS satellite's yaw attitude. This is applied only when the GPS orbit refers to the centre of mass (precise and ultra-rapid products); the broadcast orbit already refers to the antenna.

4.5 Phase wind-up

The GPS signal is circularly polarised. When the transmitting or receiving antenna rotates relative to the other, the measured phase changes by the rotation angle ω, even with no change in distance. For a fast-rotating LEO this reaches several centimetres. The program computes ω from both antennas' orientations and applies it with the IF (narrow-lane, 10.7 cm) wavelength (`PhaseWindUpModifier`).

4.6 GPS orbits and clocks

$\mathbf{r}^s$ and δt^s come from the best GPS product available **at the cut-off**, chosen per epoch in the order Final > Rapid > Ultra-rapid > Broadcast. A product counts only if it had been published by the cut-off (Finals appear about two weeks later, Rapids the next day, Ultra-rapids every six hours with their predicted half). Broadcast messages count only with a reference time at or before the cut-off. In practice the last hours of each arc use the ultra-rapid or broadcast orbits.

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5. Cycle slips and data editing

The phase equation is only useful while the ambiguity N_i stays constant. A **cycle slip** (a jump in N_i when the receiver loses lock even briefly) starts a new unknown. Each continuous stretch is an **arc** and gets its own float ambiguity. Arcs are broken at:

Test	Combination	Threshold	Detects
Loss-of-lock flag	receiver's LLI bit	set	the receiver's own detection
Data gap	—	> 60 s	lost tracking
Melbourne–Wübbena	$\mathrm{MW} = \frac{f_1 L_1 - f_2 L_2}{f_1 - f_2} - \frac{f_1 P_1 + f_2 P_2}{f_1 + f_2}$	> 2.5 wide-lane cycles (86 cm)	slips on either frequency
Geometry-free	$L_{GF} = L_1 - L_2$	jump > 1 m	gross jumps

Test	Combination	Threshold	Detects
GPS product change	—	any	discontinuity in the satellite orbit
Residual step	post-fit phase residuals	a step	equal slips on L1 and L2 (invisible to MW and GF)

The MW combination removes geometry, clocks and ionosphere, leaving the wide-lane ambiguity ($\lambda_{WL} = c/(f_1 - f_2) = 86.2$ cm) plus code noise, so a slip shows as a jump. The slip tests run on the data thinned to 10 s; the fit itself uses the phase every 60 s and the code every 30 s.

The geometry-free threshold is loose on purpose: at LEO speed the ionosphere alone moves L_{GF} by 10–30 cm in 10 s. A tight threshold flagged 460 false slips in six hours.

After each solve, observations whose residual exceeds **4 a-posteriori sigmas** are removed and the fit is repeated.

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6. Mathematics of the orbit determination

6.1 Unknowns

For a 12-hour arc (the Sentinel-3A profile) the parameter vector is

$$\mathbf{x} = [\, \underbrace{\mathbf{r}_0, \mathbf{v}_0}_{6}\ ,\ \underbrace{C_D}_{1}\ ,\ \underbrace{c\,\delta t_{r,1}, \ldots, c\,\delta t_{r,K}}_{\text{one per epoch}}\ ,\ \underbrace{B_1, \ldots, B_J}_{\text{one per arc}}\ ]$$

In the campaign the median day had about **5 000 phase observations**, **183 arcs**, and several hundred epochs, so close to **900 unknowns**. The receiver clock is estimated freely at every epoch because a spaceborne receiver clock is not stable enough to model.

6.2 Weighted least squares

The measurement model $h(\mathbf{x})$ predicts every observation y_k. The estimate minimises the weighted sum of squared residuals:

$$J(\mathbf{x}) = \sum_k \frac{(y_k - h_k(\mathbf{x}))^2}{\sigma_k^2} = (\mathbf{y} - h(\mathbf{x}))^{\top} W (\mathbf{y} - h(\mathbf{x})), \qquad W = \operatorname{diag}(1/\sigma_k^2).$$

h is non-linear (the orbit depends on $\mathbf{r}_0, \mathbf{v}_0$ through the integrated equations of motion), so it is linearised about the current estimate and solved by **Gauss–Newton**:

$$(H^{\top} W H)\, \Delta\mathbf{x} = H^{\top} W\, (\mathbf{y} - h(\mathbf{x})), \qquad \mathbf{x} \leftarrow \mathbf{x} + \Delta\mathbf{x}, \qquad H = \frac{\partial h}{\partial \mathbf{x}}.$$

6.3 The partial derivatives

Each row of H has only about eight non-zero entries:

- **Orbit (6) and C_D (1).** By the chain rule, $\partial h/\partial \mathbf{x}_0 = (\partial h/\partial \mathbf{X}(t))\,\Phi(t, t_0)$ and $\partial h/\partial C_D = (\partial h/\partial \mathbf{X}(t))\,\Psi(t)$, where $\mathbf{X} = (\mathbf{r}, \mathbf{v})$. The **state-transition matrix** $\Phi = \partial \mathbf{X}(t)/\partial \mathbf{X}(t_0)$ and the sensitivity $\Psi = \partial \mathbf{X}(t)/\partial C_D$ are integrated with the orbit from the variational equations
$$\dot{\Phi} = A(t)\,\Phi, \qquad \dot{\Psi} = A(t)\,\Psi + \frac{\partial \mathbf{F}}{\partial C_D}, \qquad A = \frac{\partial \mathbf{F}}{\partial \mathbf{X}}.$$
- **Receiver clock (1):** $\partial h/\partial(c\,\delta t_r) = 1$ for that epoch only.
- **Ambiguity (1, phase only):** $\partial L_{IF}/\partial B_j = 1$ for that arc only.

6.4 Why a sparse solver

$H^\top W H$ is about 900 × 900, but H itself is mostly zeros. Orekit's general Gauss–Newton optimiser forms $H^\top W H$ densely, which cost most of the run time. `SparseGaussNewtonOptimizer` forms the same matrix from each row's non-zero entries only. The algebra and the result are identical; only the work changes.

6.5 Convergence, weights and covariance

- **Convergence:** the iteration stops when the largest position step is below 1 mm, or the cost J changes by less than 10^{-6} of itself (Orekit's default test stalled on some days).
- **Starting weights:** $\sigma_{\text{code}} = 1$ m, $\sigma_{\text{phase}} = 1$ cm. A final solve replaces them with the post-fit scatter of each type, so the **variance factor** $\hat{\sigma}_0^2 = J/(n_{\text{obs}} - n_{\text{par}})$ is close to 1 and the covariance means something.
- **Covariance at the cut-off:** $P_0 = \hat{\sigma}_0^2\,(H^\top W H)^{-1}$.
- **Forecast covariance:** propagated with the same matrices, $P(t) = [\Phi\ \Psi]\,P_0\,[\Phi\ \Psi]^\top$.

It is worth being clear about what this covariance says: it is the uncertainty from the measurements. It does not know that the future space weather is a guess, so beyond a few hours it is optimistic.

6.6 Getting started: the initial orbit

Gauss–Newton needs a starting point within its basin of convergence. The program builds it in three steps:

1. **Single-point positioning (SPP).** At each sampled epoch (every 300 s) it solves the four unknowns $(x, y, z, c\,\delta t_r)$ from at least four code measurements, by the same least squares, rejecting epochs with a poor geometry (PDOP > 6) or a residual RMS above 30 m. Each fix is good to a few metres.
2. **Gibbs initial orbit determination** (`IodGibbs`): three position vectors on one orbit determine the velocity at the middle one, because the three are coplanar and their cross products fix the orbit's plane and shape.
3. **Position fit:** a quick Orekit batch fit of the orbit to all SPP positions (σ = 5 m).

The SPP fit was the program's whole orbit determination before this work (the "Before" run in the results). Now it only starts the carrier-phase fit.

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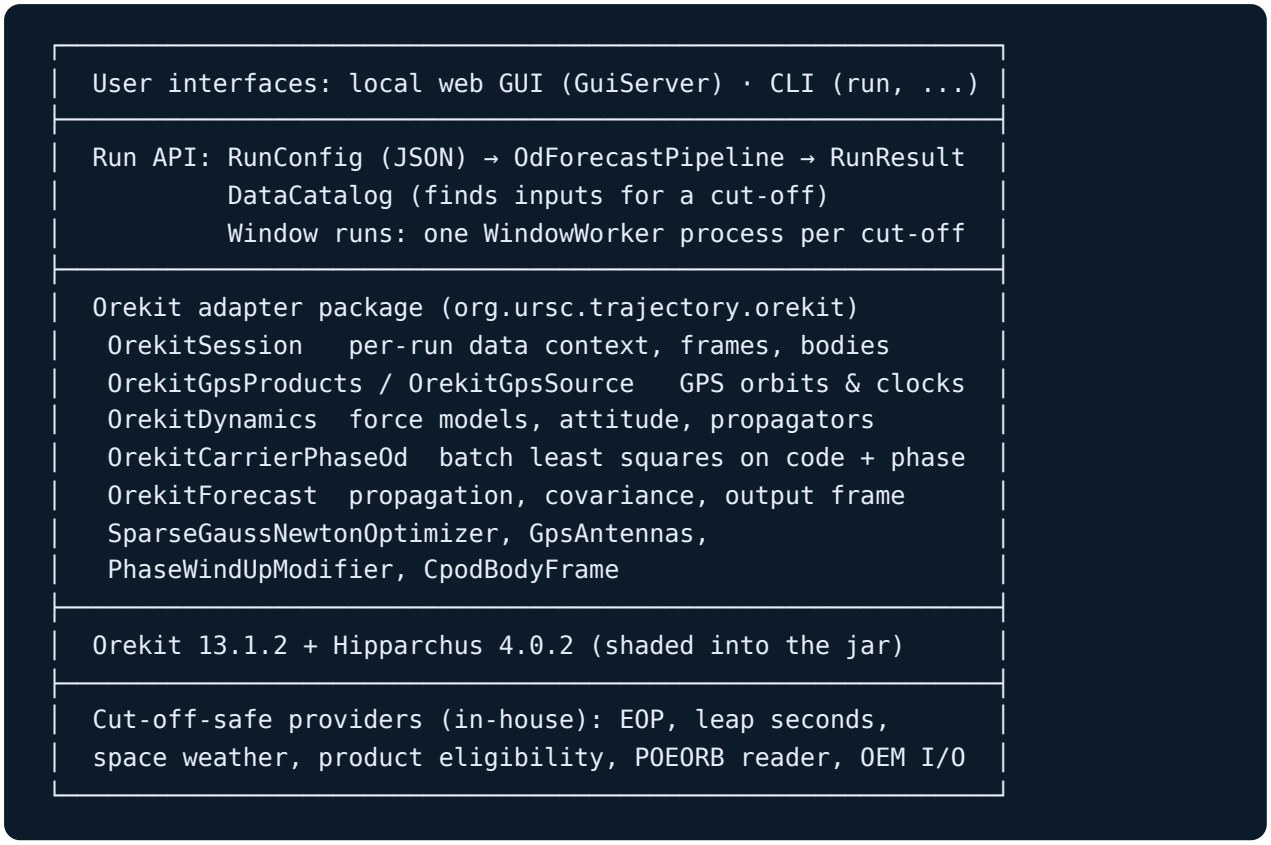
7. Mathematics of the prediction

The estimated state is integrated with Orekit's `NumericalPropagator` using a Dormand–Prince 8(5,3) integrator: an adaptive Runge–Kutta method of order 8 with an embedded order-5 error estimate. The step size is controlled so the estimated local error stays below `integrator.positionToleranceM` = 1 mm, with a maximum step of 30 s. The state is written every 60 s in the requested output frame (Orekit computes the frame transformation during the run) to a CCSDS OEM 2.0 file, optionally with the covariance of section 6.5.

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8. System design

8.1 Layers



8.2 One run, step by step

1. **Configuration.** A `RunConfig` (JSON) gives the cut-off, the spacecraft, the force switches, the OD settings and the input files. In the GUI the `DataCatalog` fills in the input files for the chosen cut-off.
2. **Orekit session.** `OrekitSession` builds a fresh Orekit data context for this run: leap seconds written from the program's table, EOP sampled daily from the cut-off-safe provider (held constant after the cut-off), EGM96, FES2004 and DE440. Nothing from one run can leak into another through Orekit's caches.

3. **Inputs.** The GNSSRD files are thinned (10 s) and parsed with Orekit's RINEX parser; the GPS products with Orekit's SP3, clock and navigation parsers, filtered by publication time.
4. **Initial orbit** (section 6.6).
5. **Carrier-phase OD** (sections 4–6). If the fit with C_D estimated fails, it is repeated with C_D fixed and a warning is added to the result (this rescued 6 and 21 September).
6. **Forecast** (section 7), and OEM output.
7. **Scoring** (only when POEORB is supplied): the forecast is compared with POEORB at every common time, as a 3-D difference and in radial / along-track / cross-track components.

8.3 Why these design choices

Choice	Reason
Orekit for all physics on the run path	a validated, maintained flight-dynamics library instead of a home-grown one
A thin adapter package around Orekit	keeps cut-off rules, configuration and GUI independent of Orekit's API
In-house providers for EOP, leap seconds, space weather	Orekit's default loaders read the newest files, which contain values from after the cut-off
One `ObservableSatellite` per epoch, one ambiguity driver per arc	works around an Orekit 13.1.2 defect: parameters split into time spans get their partial derivatives on the first span only, making the problem singular (reproducer and draft report in `docs/orekit/`)
Sparse Gauss–Newton	about 900 unknowns with 8 non-zeros per row
Separate worker process per window cut-off	Orekit's data context is process-wide; processes run cut-offs in parallel without sharing it
Single shaded jar, bundled data	runs fully offline (air-gapped)

8.4 How it differs from a pure Orekit application

The main run path calls Orekit for frames, time, forces, propagation, measurements, modifiers and the estimator. It departs from Orekit's standard pattern in these places:

Area	Standard Orekit	This program	Reason
Clocks and ambiguities	one satellite, time-span parameters	one satellite per epoch, driver per arc	Orekit 13.1.2 span defect
Optimiser	dense Gauss–Newton	sparse Gauss–Newton	speed

Area	Standard Orekit	This program	Reason
Convergence test	default	step or relative-cost test	stalls on some days
Phase wind-up	not provided for a LEO receiver	own modifier	gap in Orekit
Space weather	`CssiSpaceWeatherData`	own provider via adapter	cut-off safety
A_p input to NRLMSISE-00	daily + 3-hourly history	daily value repeated	simplification (open item)
Cycle-slip detection	Orekit detectors available	own tests	written before the migration
SPP	not in Orekit	own engine	gap in Orekit
OEM writing, POEORB reading	`OemWriter` available	own writer/reader	older code

The repository also still contains the program's earlier in-house physics library (about 150 files: forces, integrators, frames, propagators). It mirrors Orekit's structure but is used only by the CLI commands labelled "legacy".

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9. Experiments

9.1 Set-up

Item	Value
Satellite	Sentinel-3A
Period	3 September – 2 November 2024 (61 cut-offs, one per day at 05:59:42 UTC)
Data arc	6 h (fixed-C_D runs) or 12 h (final) ending at the cut-off
Forecast	7 days, state every 60 s
Reference	ESA POEORB (scoring only)
Metric	3-D position difference at the cut-off, +1 h, +6 h, +1 day, +3 days, +7 days
Statistics	median over scored days (95th percentile in brackets)

Item	Value
Validation of the estimator	closed-loop simulation (`OrekitCarrierPhaseOdTest`)

Medians are used because the five geomagnetic-storm days produce errors 50–100 times larger than a normal day and would dominate a mean. Some horizons have no POEORB coverage on a few days; those are left out ("—").

9.2 Closed-loop validation

Simulated dual-frequency data (code noise 0.3 m, phase 2 mm) with an independent light-time model, a receiver clock, random ambiguities, one wide-lane slip, one equal L1/L2 slip, a 1.66 m antenna offset, and a starting error of 30 m and 3 cm/s. The estimator recovers the orbit to **4.9 mm and 6 µm/s**, finds both slips, and gives a phase RMS of 1.9 mm with a variance factor of 1.07. The full pipeline test reaches 4 mm at the cut-off and 7 mm after one day. The mathematics and the code are therefore right; the errors on real data come from modelling.

9.3 The four configurations compared

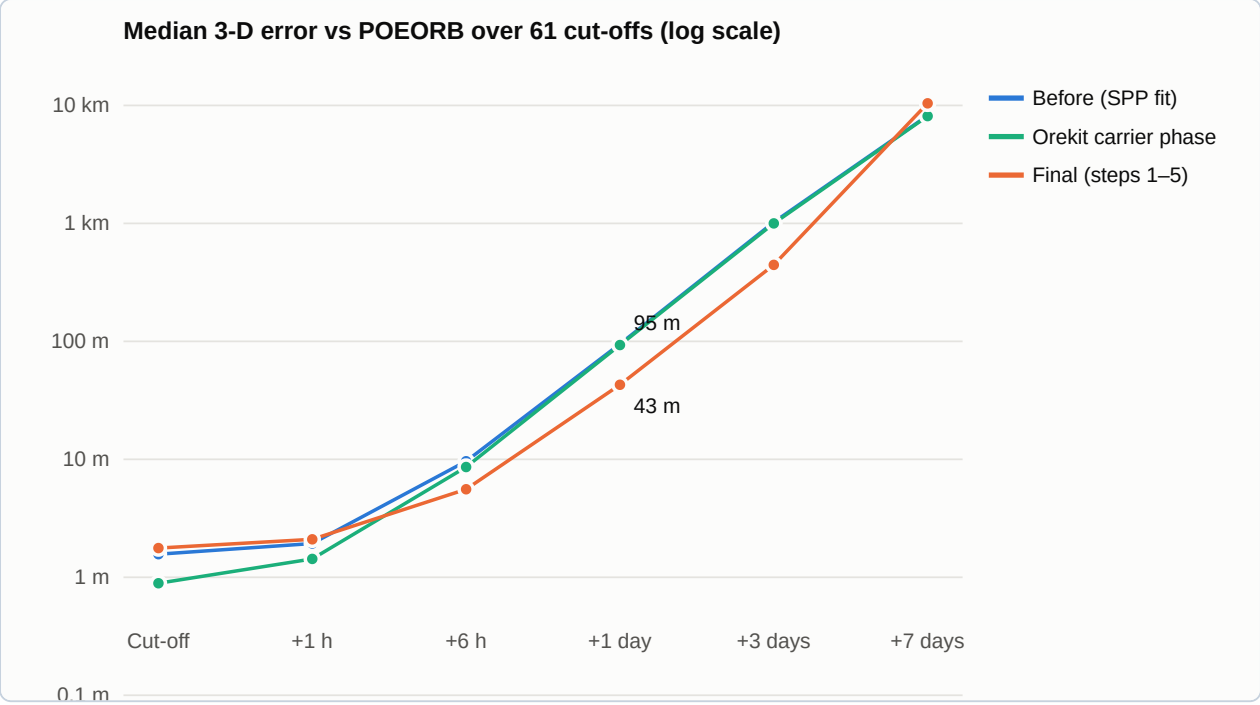
Run	Orbit determination	Engine	Arc	C_D	GPS antennas, wind-up
Before	SPP positions	in-house	6 h	fixed	no
After	carrier phase	in-house	6 h	fixed	no
Orekit	carrier phase	Orekit	6 h	fixed	no
Final	carrier phase	Orekit	**12 h**	**estimated**	**yes**

9.4 Campaign results

Median 3-D error against POEORB (m), 61 cut-offs:

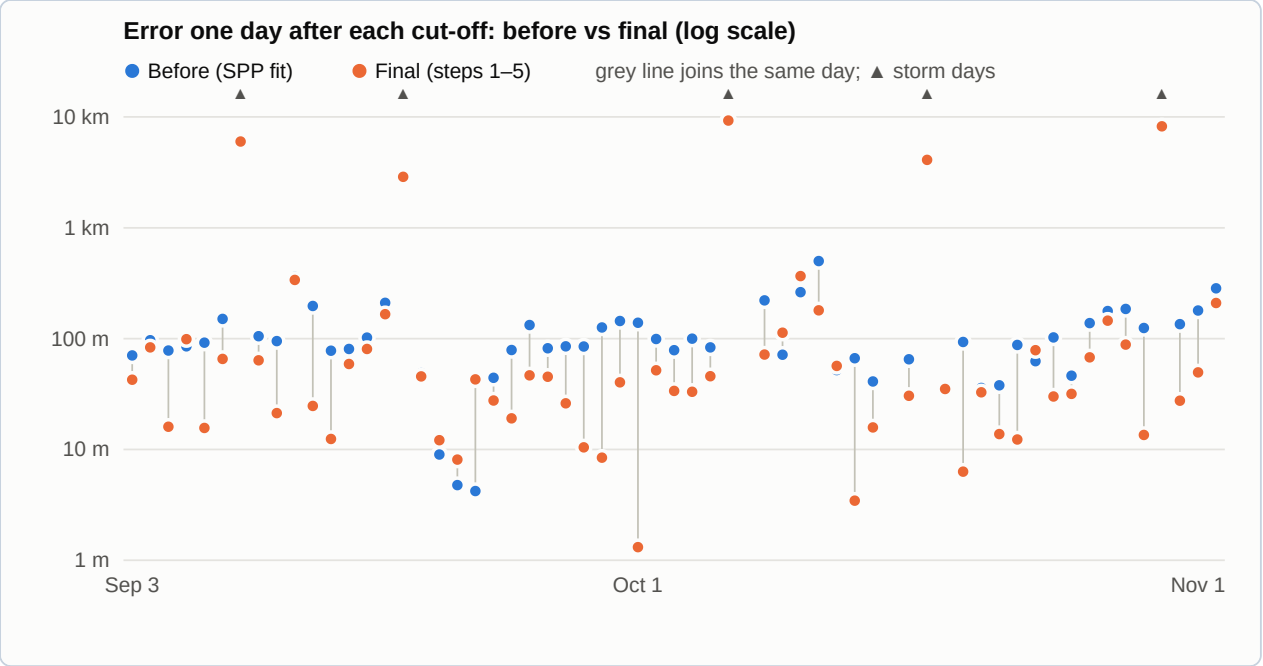
Horizon	Before	After (in-house CP)	Orekit CP	Final	Final vs Before
At cut-off	1.57 (3.21)	0.78 (2.15)	0.89 (2.22)	**1.77** (3.19)	+12 %
+1 h	1.93 (4.48)	1.26 (4.52)	1.43 (4.68)	**2.10** (4.06)	+9 %
+6 h	9.62 (712)	8.80 (713)	8.61 (714)	**5.57** (723)	−42 %
+1 day	94.91 (4,211)	92.74 (4,212)	93.06 (4,213)	**42.79** (4,286)	**−55 %**

Horizon	Before	After (in-house CP)	Orekit CP	Final	Final vs Before
+3 days	1,023 (16,355)	1,020 (16,361)	1,000 (16,404)	**445** (17,245)	**−57 %**
+7 days	8,208 (42,008)	8,205 (42,014)	8,131 (42,118)	**10,416** (47,805)	+27 %



How often Final beats Before, day by day:

Horizon	Days Final is better
At cut-off	31 of 59
+1 h	29 of 59
+6 h	40 of 59
+1 day	**46 of 59**
+3 days	37 of 58
+7 days	25 of 54



All 61 cut-offs succeeded in every run. The final run's median phase residual is 12.2 cm, the median code residual 1.26 m, with a median of 6 cycle slips per day, and a median run time of 732 s per cut-off with four running in parallel.

9.5 Step-by-step (ablation) on seven test days

Test days: 3, 11, 19 and 27 September; 6, 19 and 27 October. Median error (m):

	Orekit CP	Cd, 6 h	Cd, 12 h	+ attitude, no GPS antennas	All (final)	+ JB2008	+ box-wing
Phase RMS	6.5 cm	6.4 cm	13.0 cm	9.4 cm	12.3 cm	12.3 cm	16.8 cm
At cut-off	0.73	1.01	1.75	1.29	1.73	1.69	2.08
+6 h	10.78	5.44	4.39	4.41	4.20	7.15	22.38
+1 day	86.10	51.40	44.64	44.49	**42.50**	85.41	172
+3 days	729	513	508	507	**491**	1,045	1,509
+7 days	5,832	4,956	3,442	2,907	3,436	5,629	3,608

(The "no GPS antennas" and "box-wing" columns ran on 3 days; the others on 7.)

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10. Interpreting the results

1. Carrier phase halves the error at the cut-off. Going from SPP positions (metres of code noise) to carrier phase (centimetres) took the median from 1.57 m to 0.78–0.89 m. This is

the measurement physics of section 4 paying off.

2. The orbit-determination method alone does not change the forecast. The Before, After and Orekit runs are within 2 % of each other at +1 day and beyond. A better starting point does not help when the error after a few hours is set by drag (section 3.4).

3. Estimating C_D is what improves the forecast. It halves the equivalent acceleration error (from 8.5×10^{-9} to 3.8×10^{-9} m/s²) and therefore the error from 6 hours to 3 days: +1 day 95 → 43 m, +3 days 1.0 → 0.45 km. A longer arc (12 h instead of 6 h) separates C_D from the initial state better: +7 days 4.96 → 3.44 km on the test days.

4. The cost: a slightly worse fit at the cut-off. A purely dynamic 12-hour arc cannot follow the real orbit as closely at its end as a 6-hour arc; the phase residuals rise from about 6 to 12 cm and the cut-off error from 0.9 to 1.8 m. There are no empirical accelerations to absorb the remaining force-model error.

5. Beyond 3 days, C_D estimation can backfire. A C_D fitted over half a day of the atmosphere carries that half-day's density error into a week whose space weather is different. On 10-11 the +7-day error went from 1.5 to 24 km; on 10-07 from 8.8 km to 94 m. For 7-day products a fixed C_D remains the better choice.

6. Storms are outside any cut-off-safe method. On 9 and 18 September and 6, 17 and 30 October the +1-day error is 3–9 km in every run. A geomagnetic storm after the cut-off raised the density while the model assumed $A_p = 5$. Only a space-weather forecast could help.

7. Two drag-model experiments made things worse. JB2008 doubled the +1-day error on the test days, with its inputs held at the cut-off in the same way; at this altitude NRLMSISE-00 with an estimated C_D simply did better. The box-wing model with measured attitude tripled it, because its panel areas are generic, not Sentinel-3A's macro-model. Neither is used.

8. GPS antennas and wind-up improve the measurement model, lowering the phase residuals on the paired days (9.4 → 8.7 cm; on 3 September 6.1 → 4.9 cm), with forecast changes of a few per cent.

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11. Limitations and next steps

Limitation	Effect	Next step
Future space weather is persisted / assumed quiet	storm days 3–9 km at +1 day	use published $F_{10.7}$ / A_p forecasts as scenario inputs
Daily A_p repeated for NRLMSISE-00's 3-hourly history	weaker response to storms within the arc	pass the 3-hourly A_p

Limitation	Effect	Next step
Strictly dynamic arc, no empirical accelerations	12 cm phase residuals, 1.8 m at cut-off	estimate once-per-revolution along-track accelerations
Float ambiguities	the phase's precision is not fully used	integer ambiguity resolution (needs GPS phase biases)
Sentinel-3A macro-model and LEO antenna phase pattern unavailable	box-wing worse than cannonball	obtain the mission values
Orekit span defect	per-epoch workaround	return to standard spans once fixed upstream
Legacy in-house physics library still in the repository	two code bases to maintain	archive it

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12. Summary

- The **physics**: an orbit integrated under eleven force models, of which drag, driven by space weather that is unknown after the cut-off, dominates the forecast error; its along-track effect grows as $\frac{3}{2}\delta f t^2$.
- The **measurements**: ionosphere-free GPS code and carrier phase, with light time, Sagnac, Shapiro, relativistic clock, antenna offsets on both ends and phase wind-up modelled.
- The **mathematics**: a weighted, sparse Gauss–Newton batch least squares over about 900 unknowns (orbit, C_D , per-epoch clocks, per-arc ambiguities), with partials from the state transition matrix, outlier editing and variance re-weighting, and covariance propagated with $[\Phi\, \Psi]$.
- The **system**: an offline Java application whose physics runs on Orekit 13.1.2 through a thin adapter, with strict cut-off safety, one Orekit context per run and parallel worker processes.
- The **results** over 61 days: carrier phase halves the cut-off error; estimating C_D on a 12 h arc halves the 6-hour-to-3-day forecast error (+1 day: 95 → 43 m); a week ahead and on storm days the unknown future atmosphere sets the limit.